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| **Fiscal Year:** | FY 2022 | **Task Last Updated:** | FY 03/01/2022 |
| **PI Name:** | Rice, Kelly Ph.D. | | |
| **Project Title:** | Assessing the Impact of Agr Quorum Sensing on Staphylococcus aureus Physiology in the Space Flight Environment | | |
| **Division Name:** | Space Biology | | |
| **Program/Discipline:** | | | |
| **Program/Discipline--Element/Subdiscipline:** | | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | (1) Cell & Molecular Biology<br>(2) Microbiology | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **City:** | Gainesville | **State:** | FL |
| **Zip Code:** | 32611-0700 | **Congressional District:** | 3 |
| **Comments:** | | | |
| **Project Type:** | Flight,Ground | **Solicitation / Funding Source:** | 2018 Space Biology (ROSBio)<br>NNH18ZTT001N-FG2. App D: Flight and Ground Space Biology Research |
| **Start Date:** | 02/08/2021 | **End Date:** | 02/07/2024 |
| **No. of Post Docs:** | 0 | **No. of PhD Degrees:** | 0 |
| **No. of PhD Candidates:** | 2 | **No. of Master' Degrees:** | 0 |
| **No. of Master's Candidates:** | 0 | **No. of Bachelor's Degrees:** | 0 |
| **No. of Bachelor's Candidates:** | 1 | **Monitoring Center:** | NASA KSC |
| **Contact Monitor:** | Freeland, Denise | **Contact Phone:** | 321-867-5878 |
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| **Flight Program:** | ISS | | |
| **Flight Assignment:** | | | |
| **Key Personnel Changes/Previous PI:** | Dr. Robert Shields (Co-I) moved from the University of Florida to Arkansas State University in 2021, where he is now an Assistant Professor. He will continue to be a Co-I on this grant; UF and ASU will execute a subaward starting in year 2. | | |
| **COI Name (Institution):** | Carroll, Ronan Ph.D. ( Ohio University )<br>Edelmann, Mariola Ph.D. ( University of Florida, Gainesville )<br>Nicholson, Wayne Ph.D. ( University of Florida, Gainesville )<br>Shields, Robert Ph.D. ( University of Florida, Gainesville ) | | |
| **Grant/Contract No.:** | 80NSSC21K0601 | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

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| **Task Description:** | Effective protection of astronaut health is predicated on a thorough understanding of the effects of the space flight environment on virulence potential of bacterial pathogens. Staphylococcus aureus is a pathogen that can cause a wide variety of devastating infections, many of which result from the ability of S. aureus to grow as biofilm, its wide arsenal of secreted virulence factors, and its highly adaptive metabolism. These properties are regulated, in part, by its Accessory Gene Regulator (Agr) quorum sensing system. Analysis of S. aureus global gene expression data from a previous space flight experiment (Biological Research In Canisters (BRIC)-23; GeneLab Data System (GLDS)-145)) on the International Space Station (ISS) suggested that the Agr-mediated quorum sensing system (QSS) and expression of some of its target regulon (secreted toxins, exoproteases) were highly upregulated during space flight growth. The overall hypothesis of this proposal is that the S. aureus Agr quorum-sensing system is an important regulator of biofilm formation, physiology, and virulence factor expression when this bacterium is exposed to microgravity or other variables related to the space flight environment. To address this, S. aureus biofilm formation in a wildtype and isogenic agr-deficient mutant strain will be compared using a random positioning machine (RPM) to simulate microgravity, and an ISS space flight experiment will be conducted using 4 Biological Research in Canister Petri Dish Fixation Units (BRIC-PDFUs). One BRIC-PDFU (n=5 PDFUs + 1 HOBO temperature logger) will be used to grow a S. aureus transposon library, in order to define which gene(s) are required for thriving in the space flight environment. Two BRIC-PDFUs will be used to grow S. aureus wildtype (same strain used in BRIC-23) and its isogenic agr mutant (n=5 PDFUs + 1 HOBO temperature logger per strain), to determine the contribution of Agr quorum sensing to gene expression changes previously observed in the BRIC-23 flight samples. In a fourth BRIC, PDFUs modified to cancel the potential confounding effects of microgravity on fluid geometry, and hence aeration, in BRIC units (modifications are being developed by Co-Investigator Nicholson through a separate NASA project), will be used to grow the wild-type S. aureus strain (n=5 modified PDFUs + 1 HOBO temperature). The growth media, time, and other experimental variables will be identical to those previously used in the BRIC-23 experiment in order to directly compare results. Asynchronous ground controls for each of the four space flight BRIC-PDFUs (n=5 PDFUs + 1 HOBO temperature logger per BRIC) described above will be treated identically in the ISS Environmental Simulator at Kennedy Space Center (KSC) using the recorded flight temperature data. Upon return to Earth, flight samples and ground-based controls for the wild-type and agr mutant BRIC-PDFUs will be processed for RNA-seq, secreted proteomics, metabolite, and growth measurements. The transposon library BRIC-PDFUs will be processed for DNA isolation, Illumina sequencing, and Tn-seq analysis. These proposed experiments will allow us to make more refined predictions about the pathogenic potential of S. aureus, especially in relation to astronaut health. These experiments align with Decadal Survey Recommendation P1, sub-topics MB-B (Studies designed to determine how spaceflight conditions affect microbial biofilm formation... and their potential to cause disease) and MB-D (to use the ISS as a microbial observatory to determine foundational results... that contribute to the health of future space explorers), as well as address several guiding questions of the 2016-2025 SB Science Plan: To help identify the "underlying genetic, molecular and biochemical processes influenced by the space environment" for S. aureus (MB-1), and offer additional insight into how space flight conditions such as microgravity "...influence microbial biofilm formation and microbial community development" (MB-3). |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | Agr is a global regulator of both virulence and metabolic genes, therefore stimulation of this quorum-sensing system in response to space flight conditions may impact the pathogenic threat of S. aureus to astronauts during long-term space flight missions, especially since space flight is known to impose a suppressive effect on both innate and adaptive immune cell responses to pathogen-related stimuli. The research proposed in this application is directed at gaining a better understanding of the contribution of Agr quorum sensing to physiological adaptation and pathogenic potential of S. aureus in response to the space flight environment. It is anticipated that the results of these proposed experiments will allow us to make more refined predictions as to which virulence properties and physiological changes are likely to occur in S. aureus during space flight, and how this may impact its pathogenic potential and effect on astronaut health. |
| **Task Progress:** | Aim 1: ISS flight experiment to elucidate the physiological response of S. aureus to the space flight environment<br>• In year 1, we have focused on collecting baseline data to determine the effects of culture media composition on S. aureus growth and agr expression, to best optimize the media composition to use for pre-flight BRIC-PDFU verification testing for S. aureus UAMS-1 and isogenic agr mutants, which will be conducted in year 2. Based on these results, we decided to continue to use TSYG medium (TSYG) for future BRIC-PDFU pre-science and science verification testing and eventual flight experiment, as this will allow more direct comparison to the BRIC-23 results.<br><br>• We constructed an unmarked agr deletion mutant in UAMS-1 to use in these studies. Dr. Wayne Nicholson (Co-I) tested both the UAMS-1 wild-type and agr deletion mutant for longevity of air-dried samples of these strains deposited on petri dishes. These results confirmed earlier findings that both the wild-type and agr mutant strain retain full viability for about 14 days after air-drying on petri dishes.<br><br>• Currently we are finalizing a workflow to optimize the amount of culture volume that will be needed to harvest from each PDFU to perform RNAseq, proteomics, secretomics, and metabolomics on wildtype and agr mutant cultures grown under BRIC-like conditions (low-oxygen growth, room temperature for 48 hours).<br><br>• We have received the tools to generate a TnSeq library in UAMS-1, and also aliquots of this library already prepared in other S. aureus strain backgrounds (as a backup in case it is too difficult to generate the library in UAMS-1). In year 2, we will construct the UAMS-1 library, test it for longevity of air-dried samples on petri dishes, and perform a preliminary TnSeq experiment under BRIC-like conditions (low-oxygen growth, room temperature for 48 hours).<br><br>Aim 2: Determine the effect of Agr and simulated microgravity on S. aureus biofilm growth<br><br>• In year 1, most of the focus has been on Aim 1 objectives. However, we have performed some pilot imaging of S. aureus lab strain biofilms using the random positioning machine (RPM) model. We are currently constructing new constitutive green fluorescence protein (GFP) reporter plasmids for use in strain UAMS-1 and its agr deletion mutant, as the original construct did not express well in UAMS-1.<br><br>• We are also currently investigating the role of aeration in regulating S. aureus Agr and metabolism using the low-shear modeled microgravity (LSMMG) model of growth. |

| Bibliography Type: | Description: (Last Updated: 02/19/2026) |
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| Abstracts for Journals and Proceedings | Hauserman MR, Edelmann MJ, Carroll RK, Rice KC. "Staphylococcus aureus quorum sensing response is affected by spaceflight and media conditions." 37th Annual Meeting of the American Society for Gravitational and Space Research, Baltimore, MD, November 3-6, 2021.<br>Abstracts. 37th Annual Meeting of the American Society for Gravitational and Space Research, Baltimore, MD, November 3-6, 2021. , Nov-2021 |
| Abstracts for Journals and Proceedings | Hauserman MR, Carroll RK, Edelmann MJ, Rice KC. "Analysis of BRIC-23 GeneLab process verification test multi-omics data reveals altered quorum sensing and virulence potential of staphylococcus aureus during space flight." 37th Annual Meeting of the American Society for Gravitational and Space Research, Baltimore, MD, November 3-6, 2021.<br>Abstracts. 37th Annual Meeting of the American Society for Gravitational and Space Research, Baltimore, MD, November 3-6, 2021. , Nov-2021 |